

1. Numpy. Numpy merupakan library python yang digunakan untuk KOMPUTASI MATRIKS
2. Matplotlib.pyplot merupakan library python untuk PRESENTASI GRAFIK ATAU PLOT
3. Pandas. merupakan library python untuk MENGIMPOR EKSTENSI DATA DARI LUAR (CSV, EXCEL atau TEXT)

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

1. Bagaimana kita bisa membaca dataset .CSVnya? dengan menuliskan "dataset = pd.read_csv", kemudian kita masukkan variabel x dan y Variabel X, Atribut variabel Y, Label
2. Atribut dan Label dibagi menjadi 2 Variabel, yaitu Atribut (x) dan Label (y)
3. .iloc adalah fungsi untuk memilih (subsetting) baris atau kolom dari sebuah dataset/dataframe berdasarkan posisi indeks numerik(angka), bukan berdasarkan nama label kolom atau barisanya.
4. .values digunakan untuk mengambil data mentah di dalam dataset/dataframe dan mengubahnya menjadi Array NumPy. Saat menggunakan .values, semua elemen visual seperti nama label baris

```
dataset = pd.read_csv('/content/Dataset_Missing_Values.csv')
x = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
```

```
print (x)
```

```
[['France' 44.0 72000.0]
 ['Spain' 27.0 48000.0]
 ['Germany' 30.0 54000.0]
 ['Spain' 38.0 61000.0]
 ['Germany' 40.0 nan]
 ['France' 35.0 58000.0]
 ['Spain' nan 52000.0]
 ['France' 48.0 79000.0]
 ['Germany' 50.0 83000.0]
 ['France' 37.0 67000.0]]
```

```
print (y)
```

```
['No' 'Yes' 'No' 'No' 'Yes' 'Yes' 'No' 'Yes' 'No' 'Yes']
```

1. Setelah berhasil mengimport data, selanjutnya kita akan mengisikan "Missing values" = NAN. caranya bagaimana?
2. Untuk mengisikan missing values kita dapat menggunakan function "SIMPLE IMPUTER", untuk memanggil function IMPUTER yang nantinya untuk menggunakan nilai "RATA RATA ATAU MEAN" dari kolomnya menggunakan library "SKLEARN.IMPUTE", kemudian kita panggil functionnya yaitu SIMPLE IMPUTER.

```
from sklearn.impute import SimpleImputer
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
imputer.fit(x[:, 1:3])
x[:, 1:3] = imputer.transform(x[:, 1:3])
```

```
print (x)
```

```
[['France' 44.0 72000.0]
 ['Spain' 27.0 48000.0]
 ['Germany' 30.0 54000.0]
 ['Spain' 38.0 61000.0]
 ['Germany' 40.0 63777.77777777778]
 ['France' 35.0 58000.0]
 ['Spain' 38.77777777777778 52000.0]
 ['France' 48.0 79000.0]
 ['Germany' 50.0 83000.0]
 ['France' 37.0 67000.0]]
```

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
ct = ColumnTransformer(transformers=[('encoder', OneHotEncoder(), [0])], remainder='passthrough')
x = np.array(ct.fit_transform(x))
```

```
print (x)
```

```
[[1.0 0.0 0.0 44.0 72000.0]
 [0.0 0.0 1.0 27.0 48000.0]
 [0.0 1.0 0.0 30.0 54000.0]
 [0.0 0.0 1.0 38.0 61000.0]
 [0.0 1.0 0.0 40.0 63777.77777777778]]
```

```
[1.0 0.0 0.0 35.0 58000.0]
[0.0 0.0 1.0 38.77777777777778 52000.0]
[1.0 0.0 0.0 48.0 79000.0]
[0.0 1.0 0.0 50.0 83000.0]
[1.0 0.0 0.0 37.0 67000.0]]
```

Encoding data kategori (Class / Label)

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
y = le.fit_transform(y)
```

```
print(y)
```

```
[0 1 0 0 1 1 0 1 0 1]
```

Membagi dataset ke dalam training set dan test set

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size = 0.2, random_state = 1)
```

```
print(x_train)
```

```
[[0.0 0.0 1.0 38.77777777777778 52000.0]
 [0.0 1.0 0.0 40.0 63777.77777777778]
 [1.0 0.0 0.0 44.0 72000.0]
 [0.0 0.0 1.0 38.0 61000.0]
 [0.0 0.0 1.0 27.0 48000.0]
 [1.0 0.0 0.0 48.0 79000.0]
 [0.0 1.0 0.0 50.0 83000.0]
 [1.0 0.0 0.0 35.0 58000.0]]
```

```
print(x_test)
```

```
[[0.0 1.0 0.0 30.0 54000.0]
 [1.0 0.0 0.0 37.0 67000.0]]
```

```
print(y_train)
```

```
[0 1 0 0 1 1 0 1]
```

```
print(y_test)
```

```
[0 1]
```

Feature Scaling

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train[:, 3:] = sc.fit_transform(x_train[:, 3:])
x_test[:, 3:] = sc.transform(x_test[:, 3:])
```

```
print(x_train)
```

```
[[0.0 0.0 1.0 -0.19159184384578545 -1.0781259408412425]
 [0.0 1.0 0.0 -0.014117293757057777 -0.07013167641635372]
 [1.0 0.0 0.0 0.566708506533324 0.633562432710455]
 [0.0 0.0 1.0 -0.30453019390224867 -0.30786617274297867]
 [0.0 0.0 1.0 -1.9018011447007988 -1.420463615551582]
 [1.0 0.0 0.0 1.1475343068237058 1.232653363453549]
 [0.0 1.0 0.0 1.4379472069688968 1.5749910381638885]
 [1.0 0.0 0.0 -0.7401495441200351 -0.5646194287757332]]
```

```
print(x_test)
```

```
[[0.0 1.0 0.0 -1.4661817944830124 -0.9069571034860727]
 [1.0 0.0 0.0 -0.44973664397484414 0.2056403393225306]]
```

